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1.PROBLEM STATEMENT:

Real estate seekers in India often navigate through multiple property listing websites such as NoBroker, 99Acres, and Housing.com to find a suitable home or rental space. Each platform offers its own set of listings, filters, and user interfaces, leading to a fragmented and inefficient search experience. This process becomes tedious and time-consuming for users, as they must repetitively apply filters, compare listings manually, and track favorites across different platforms.

To address this issue, the OneRoof project aims to aggregate listings from multiple real estate platforms into a single, cohesive web application. This unified platform streamlines the property search process, enabling users to access a comprehensive pool of listings in one place, apply consistent filters, and interact with an intuitive user interface.

2.OBJECTIVE:

The main objective of the OneRoof project is to enhance the property searching experience by creating a centralized platform that consolidates listings from various real estate websites. The project aims to:

- Minimize the effort users spend switching between platforms.

- Ensure real-time updates and data accuracy using APIs and scraping techniques.
- Provide a clean and uniform user interface using modern frontend tools.
- Implement secure and scalable user authentication using Supabase.
- Deliver high performance through optimized backend operations using Python and MERN stack.

3.MOTIVATION:

The Indian real estate market is vast and growing, with millions of users turning to online platforms for buying, selling, or renting properties. However, the need to explore multiple platforms often causes frustration and delays. This redundancy inspired the development of OneRooF—a one-stop solution to simplify and accelerate property discovery.

By leveraging modern technologies like React, Python, and Supabase, OneRooF provides a seamless and reliable experience. Its goal is not just to save time, but to enable users to make better and faster real estate decisions using a comprehensive and intuitive tool.

4.DATA SOURCE:

The data utilized in OneRooF comes from the following sources:

- Real-time API access or web scraping of listing data from NoBroker, 99Acres, and Housing.com.
- Scraped data includes key listing attributes like location, price, configuration (1BHK, 2BHK, etc.), property type, amenities, and contact information.
- User credentials and login data are stored and verified through Supabase, which offers an open-source backend with integrated authentication.

Data is regularly updated and stored in structured formats (JSON, MongoDB) to ensure the listings remain fresh and accurate.

5. TECHNOLOGICAL BACKGROUND:

To build the OneRooF platform, the project integrates various modern technologies:

- Python: Used for building APIs and scripts to extract and parse data from real estate websites.

- React.js: Handles the dynamic and responsive frontend user interface with components and state management.
- Vite: A build tool that offers fast refresh and bundling for React development, ensuring optimal performance.
- Supabase: Provides backend services including authentication, database (PostgreSQL), and API access.
- MERN Stack (MongoDB, Express.js, React.js, Node.js): Ensures fullstack capability with scalable architecture and fast data access.

These technologies work in synergy to ensure reliability, performance, scalability, and ease of use.

6. TOOL/IMPLEMENTATION:

The OneRooF platform was implemented using the following steps:

1. Frontend Design: Developed with React.js using Vite for speed, the UI includes search filters, result cards, and detail pages. Styling is done with Tailwind CSS.
2. Backend API Integration: Built using Python scripts and Express.js, the backend fetches listings via APIs and web scraping. These are cleaned and served to the frontend.

3. Authentication: User registration and login are managed through Supabase's auth service, allowing secure sessions and real-time sync.
4. Database Management: MongoDB stores the fetched listings, while Supabase handles user data.
5. Deployment: The complete application is deployed on Vercel for global access

7. VALIDATION OR SIMULATION:

The platform was tested and validated through:

- Manual comparison of aggregated listings with original data on NoBroker, 99Acres, and Housing.com.
- Testing login and signup processes with Supabase to ensure secure authentication.
- Load testing to evaluate performance under concurrent user scenarios.
- User testing with peers to gather feedback on usability, speed, and design.

The system was iteratively improved based on testing results to optimize performance and user experience.

8. RESULTS AND CONCLUSION:

Results:

- Successfully created a unified real estate aggregator combining multiple platforms into one application.
- Implemented secure and real-time login/auth using Supabase.
- Fast, mobile-friendly frontend UI with Vite + React.js.

Conclusion:

OneRooF significantly improves the real estate browsing experience by reducing redundancy and bringing everything under one digital roof. Its modular design and scalable tech stack allow it to expand with more features, like AI-driven recommendations, property comparisons, and direct messaging with owners in the future.

9. REFERENCES:

- Supabase documentation: <https://supabase.com/docs>
- Vite official site: <https://vitejs.dev/>
- MongoDB, Express, React, Node documentation

- Stack Overflow community threads
- Python scraping libraries (BeautifulSoup, Requests)
- OneRoof Deployment: <https://onerooffinal.vercel.app>

OUR WEBSITE :

<https://one-roof-final.vercel.app/>